Project

In [15]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings("ignore")
```

In [17]:
```
df=pd.read_csv(r"C:\Users\HP\Desktop\heart2.csv")
df
```

Out[17]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3

1025 rows × 14 columns

EDA

In [18]:
```
df.shape
```

Out[18]:
```
(1025, 14)
```

In [19]:
```
df.head()
```

Out[19]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	t
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	

In [20]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1025 non-null   int64  
 1   sex       1025 non-null   int64  
 2   cp        1025 non-null   int64  
 3   trestbps  1025 non-null   int64  
 4   chol      1025 non-null   int64  
 5   fbs       1025 non-null   int64  
 6   restecg   1025 non-null   int64  
 7   thalach   1025 non-null   int64  
 8   exang     1025 non-null   int64  
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64  
 11  ca        1025 non-null   int64  
 12  thal      1025 non-null   int64  
 13  target    1025 non-null   int64  
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

In [21]:
```
df.describe()
```

Out[21]:

	age	sex	cp	trestbps	chol	fbs	r
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.00000	1025.000000	1025.0
mean	54.434146	0.695610	0.942439	131.611707	246.00000	0.149268	0.5
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.5
min	29.000000	0.000000	0.000000	94.000000	126.00000	0.000000	0.0
25%	48.000000	0.000000	0.000000	120.000000	211.00000	0.000000	0.0
50%	56.000000	1.000000	1.000000	130.000000	240.00000	0.000000	1.0
75%	61.000000	1.000000	2.000000	140.000000	275.00000	0.000000	1.0
max	77.000000	1.000000	3.000000	200.000000	564.00000	1.000000	2.0

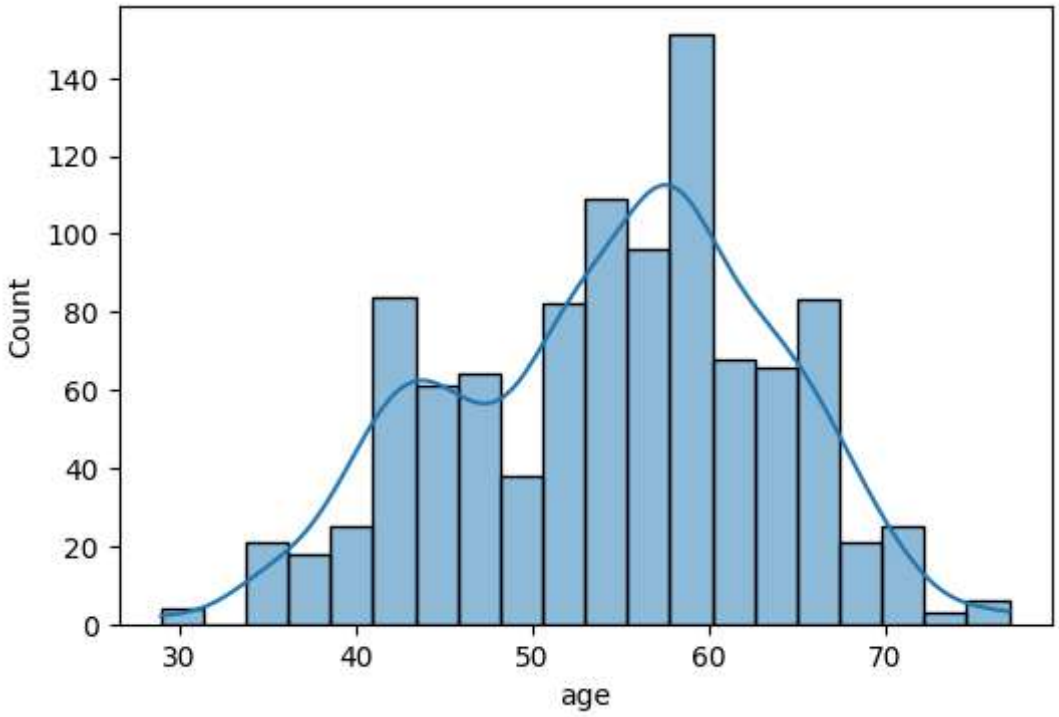
In [22]:
```
df.isnull().sum()
```

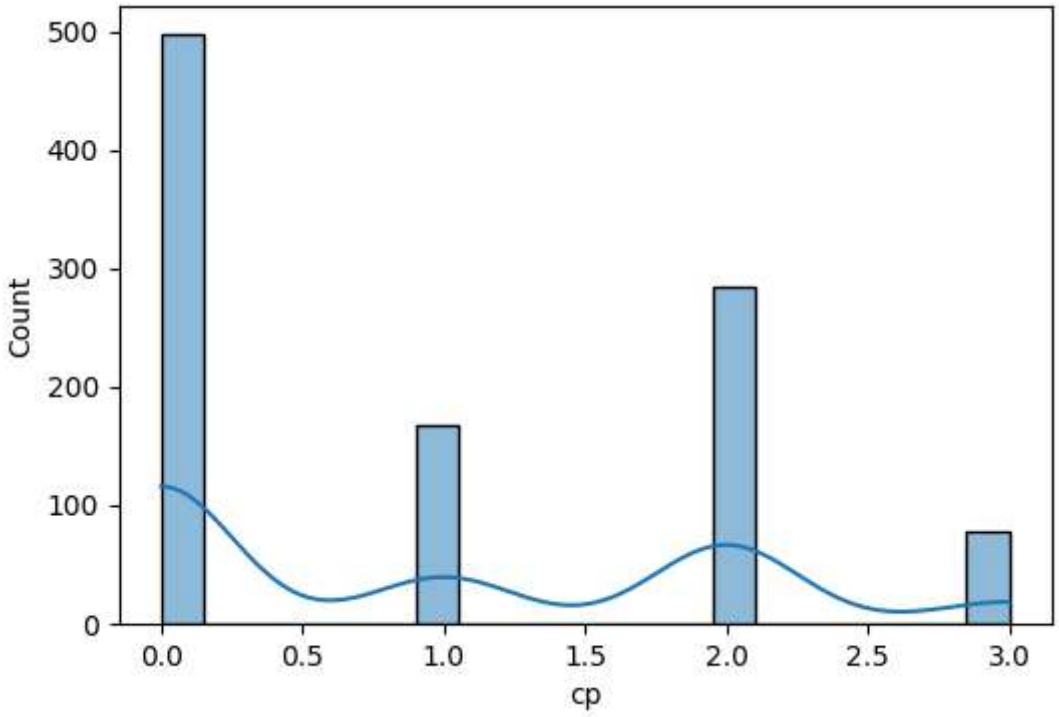
Out[22]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

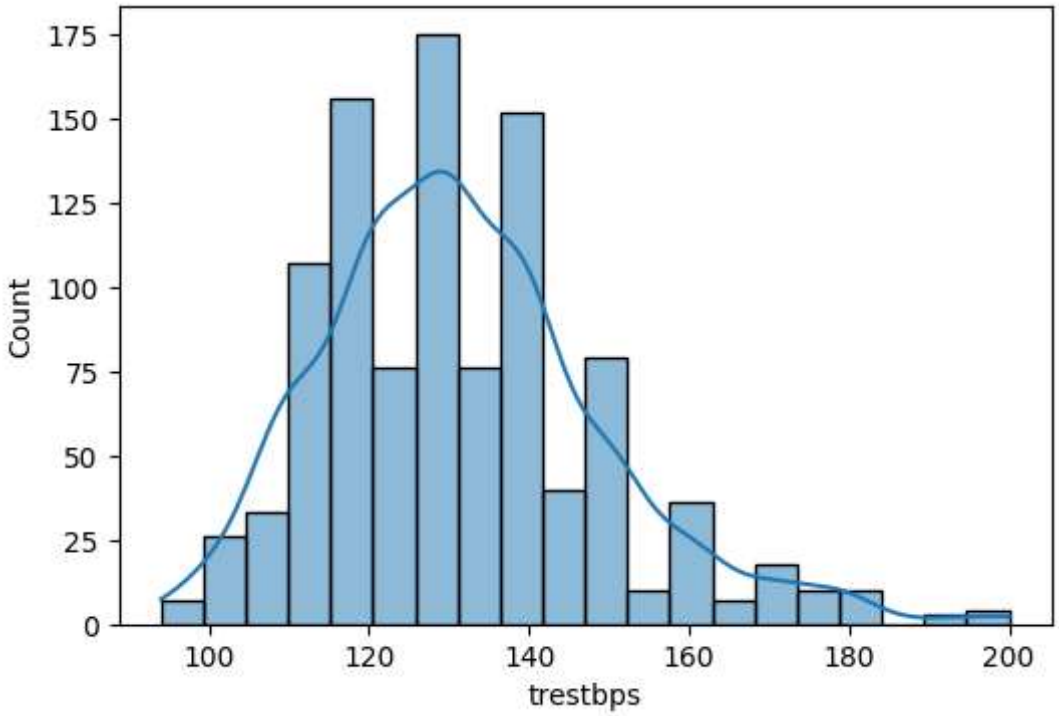
In [23]:
```
df.columns
```

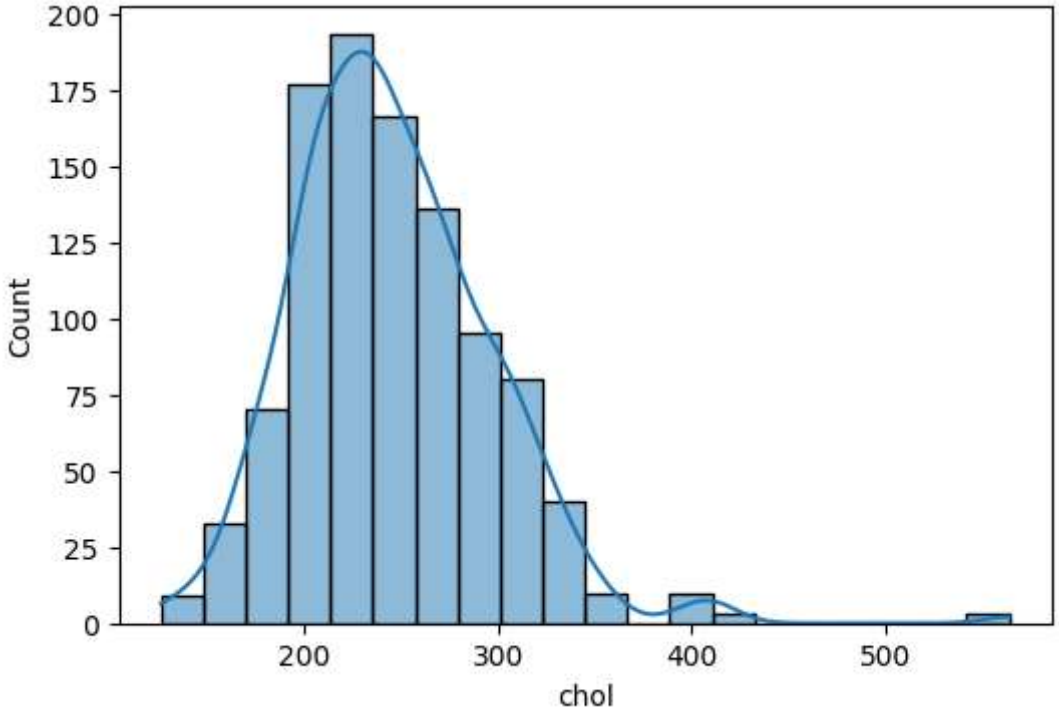
Out[23]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='str')
```

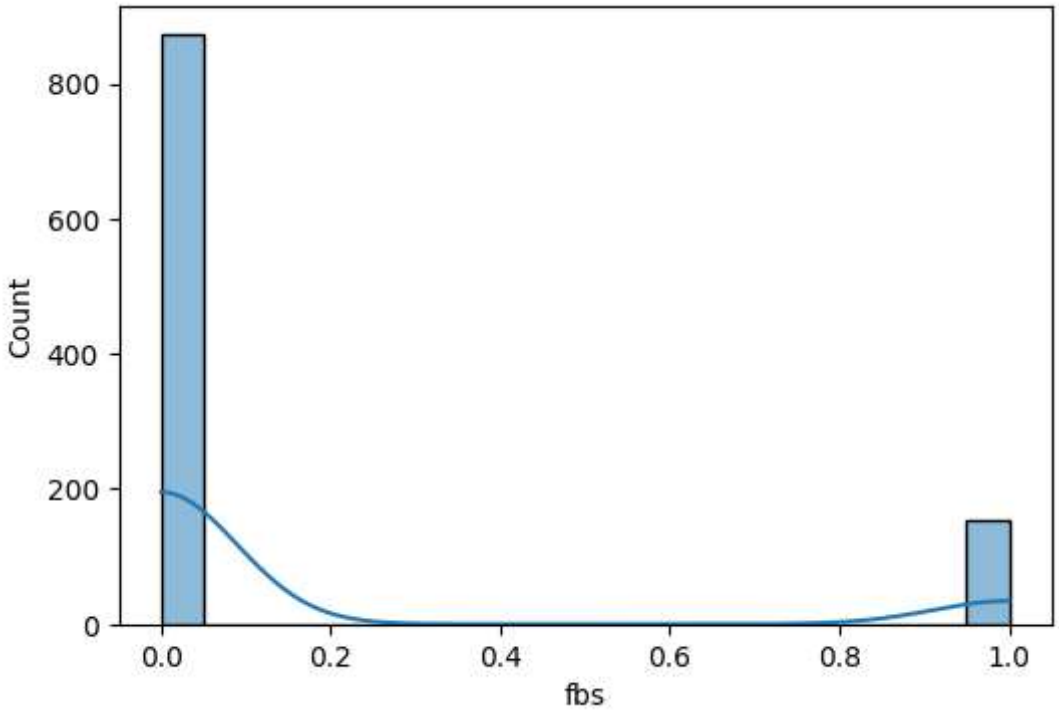
In [30]:
```
numeric_columns=["age","cp","trestbps","chol","fbs","restecg","thalach","exang","ol
for col in numeric_columns:
    plt.figure(figsize=(6,4))
    sns.histplot(df[col],kde=True,bins=20)
```

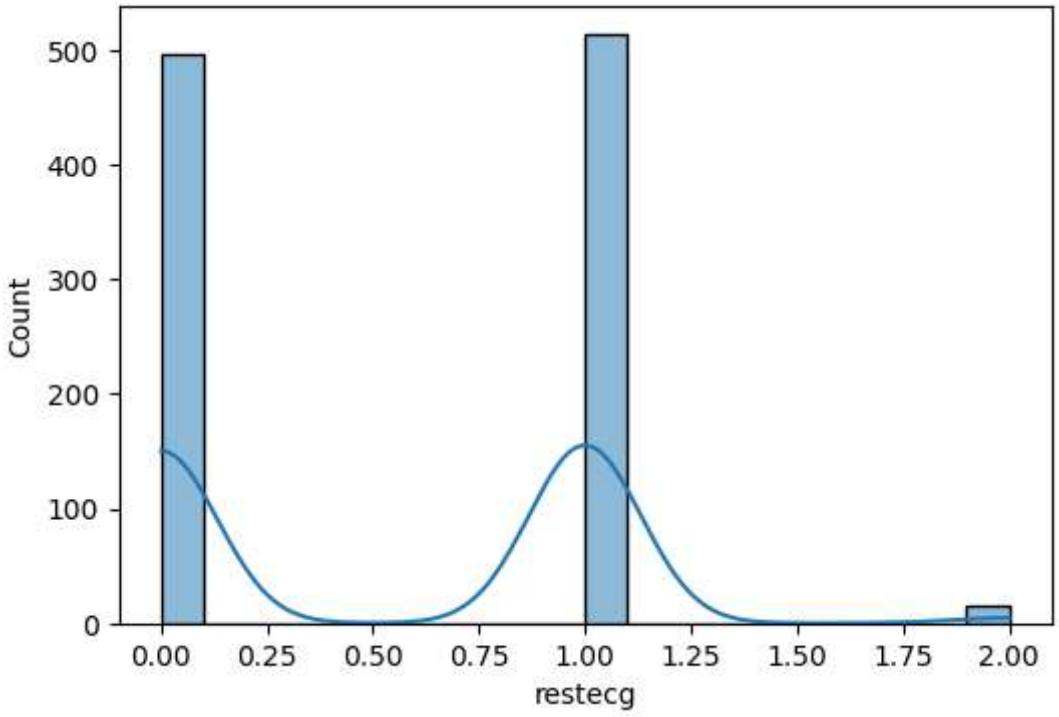

Count
140
120
100
80
60
40
20
0
30
40
50
60
70
age

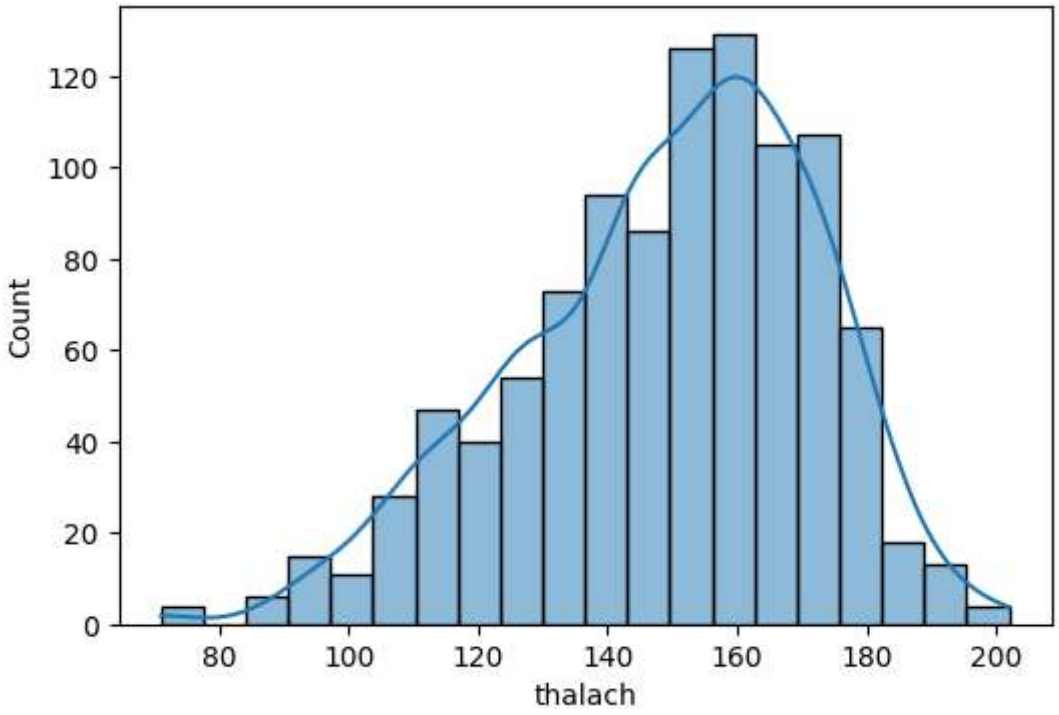

Count
500
400
300
200
100
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
cp

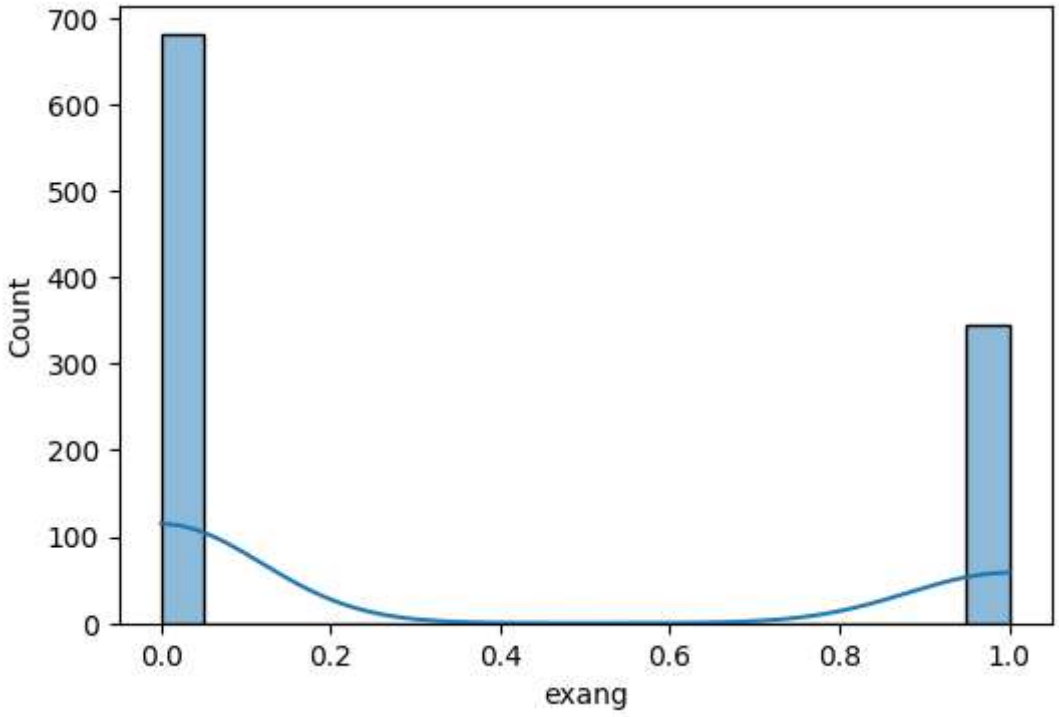

Count
trestbps
0
25
50
75
100
125
150
175
100
120
140
160
180
200

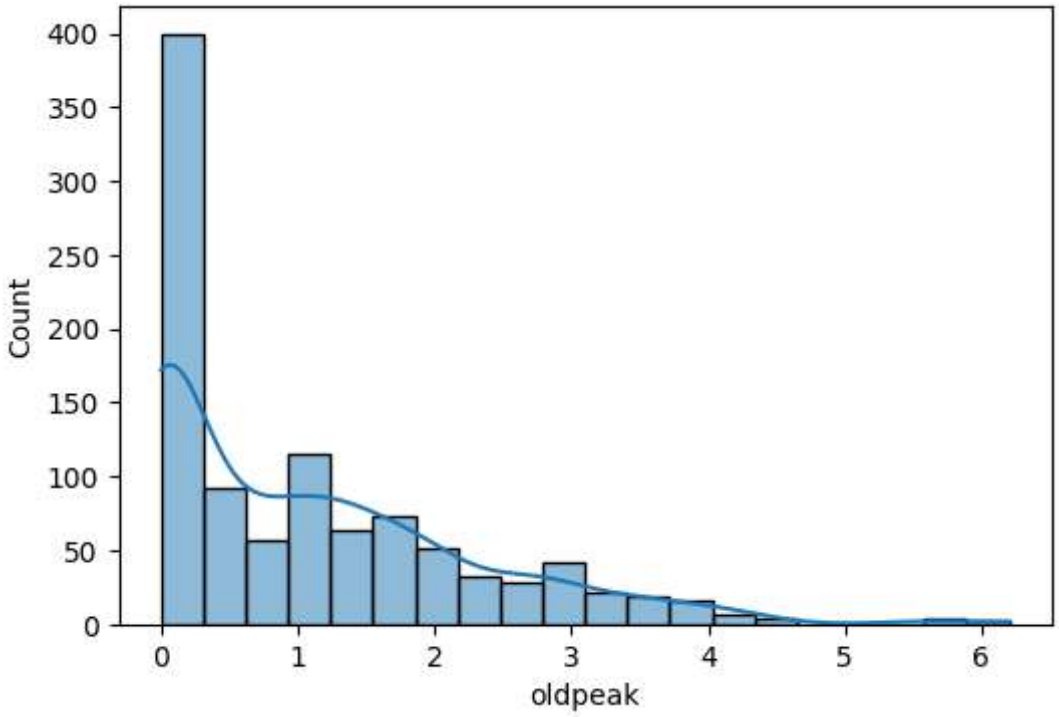

Count
chol
0
25
50
75
100
125
150
175
200
200
300
400
500

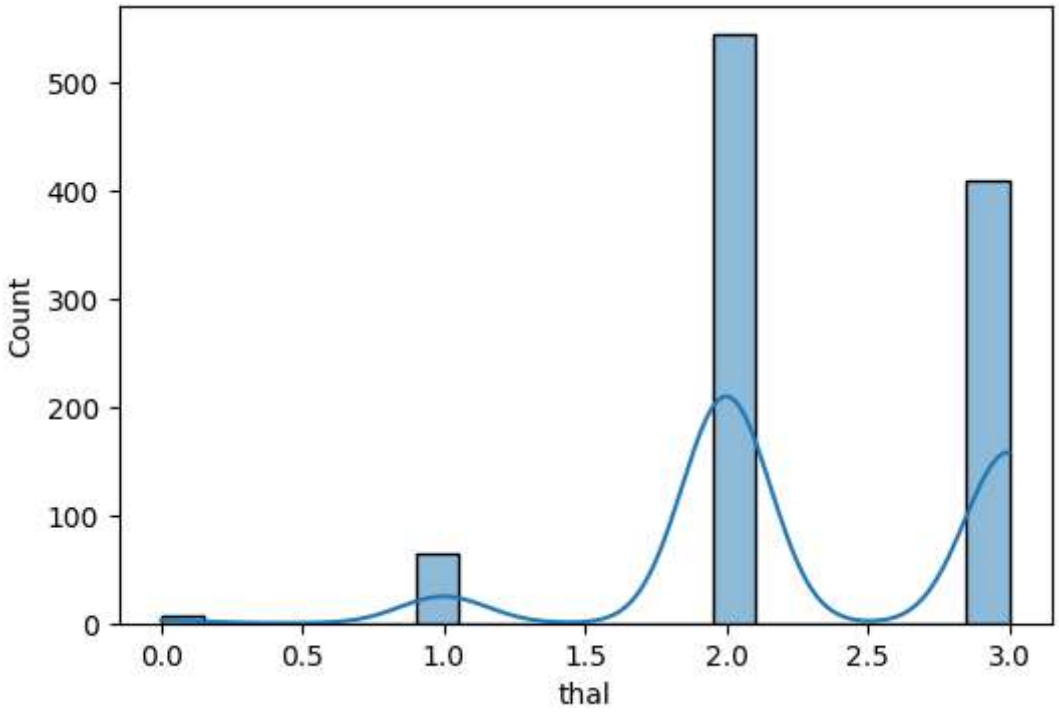

800
600
400
200
0
Count
0.0
0.2
0.4
0.6
0.8
1.0
fbs

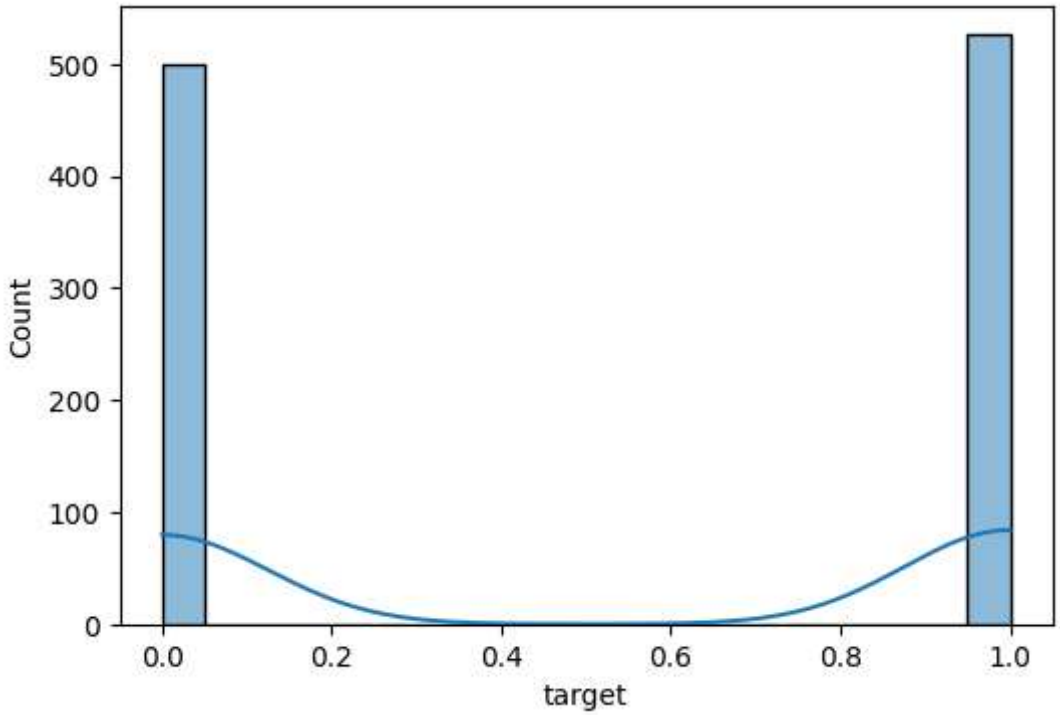

500
400
300
200
100
0
Count
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
restecg


Count
thalach
0
20
40
60
80
100
120
80
100
120
140
160
180
200


Count
exang
0
100
200
300
400
500
600
700
0.0
0.2
0.4
0.6
0.8
1.0

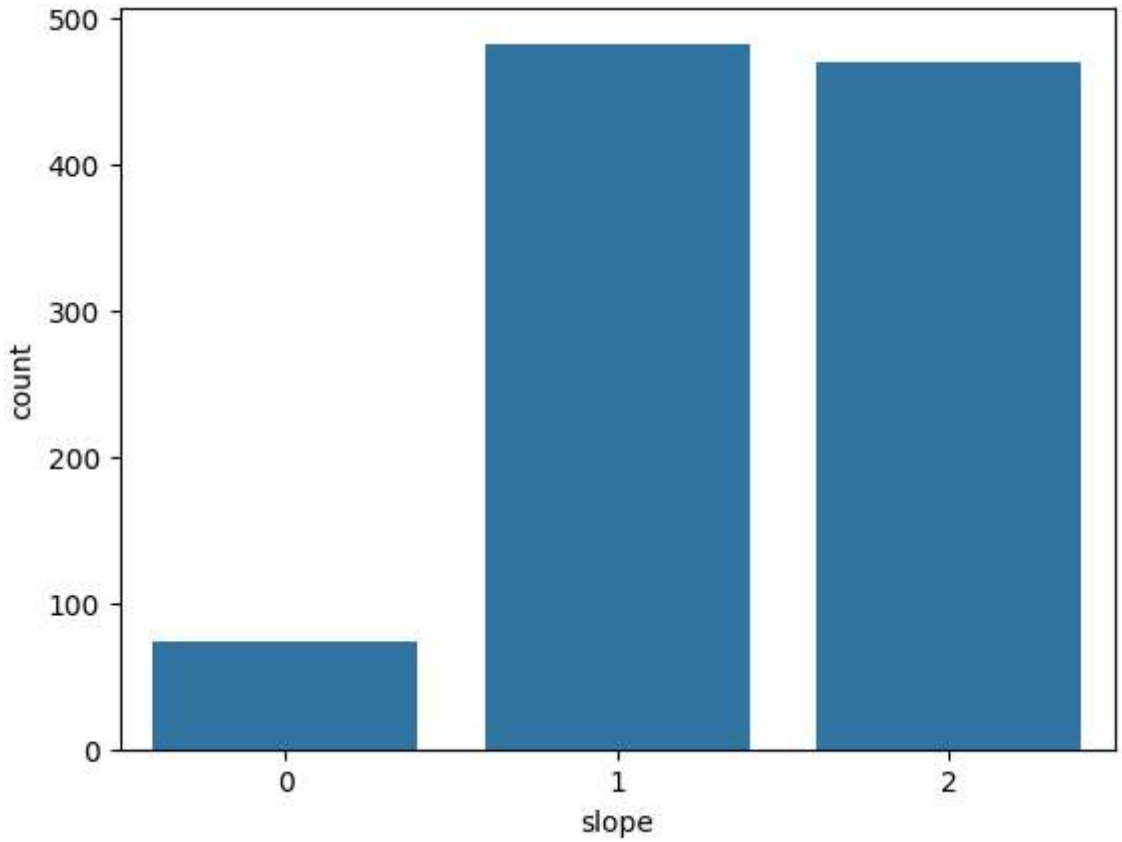

Count
oldpeak
0
50
100
150
200
250
300
350
400
0
1
2
3
4
5
6


Count
thal
0
100
200
300
400
500
0.0
0.5
1.0
1.5
2.0
2.5
3.0



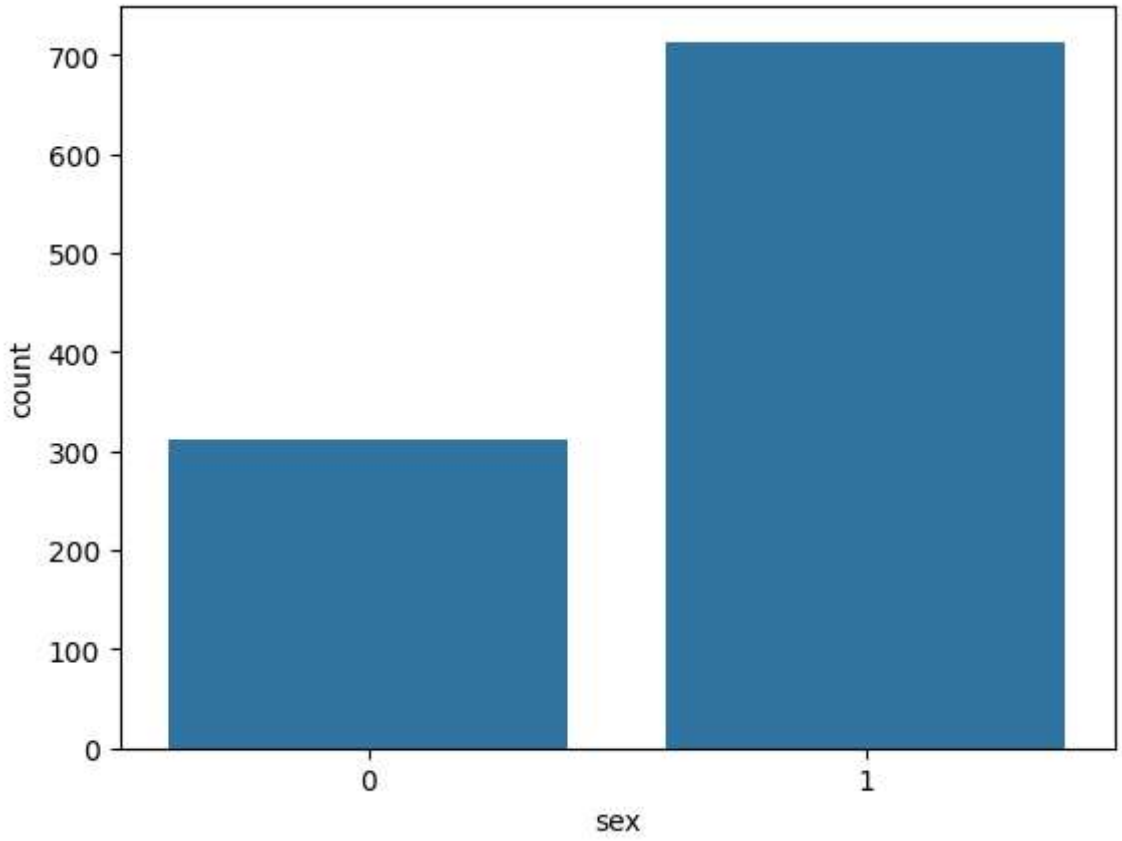
In [27]:
```
sns.countplot(x=df["slope"])
```

Out[27]:
```
<Axes: xlabel='slope', ylabel='count'>
```



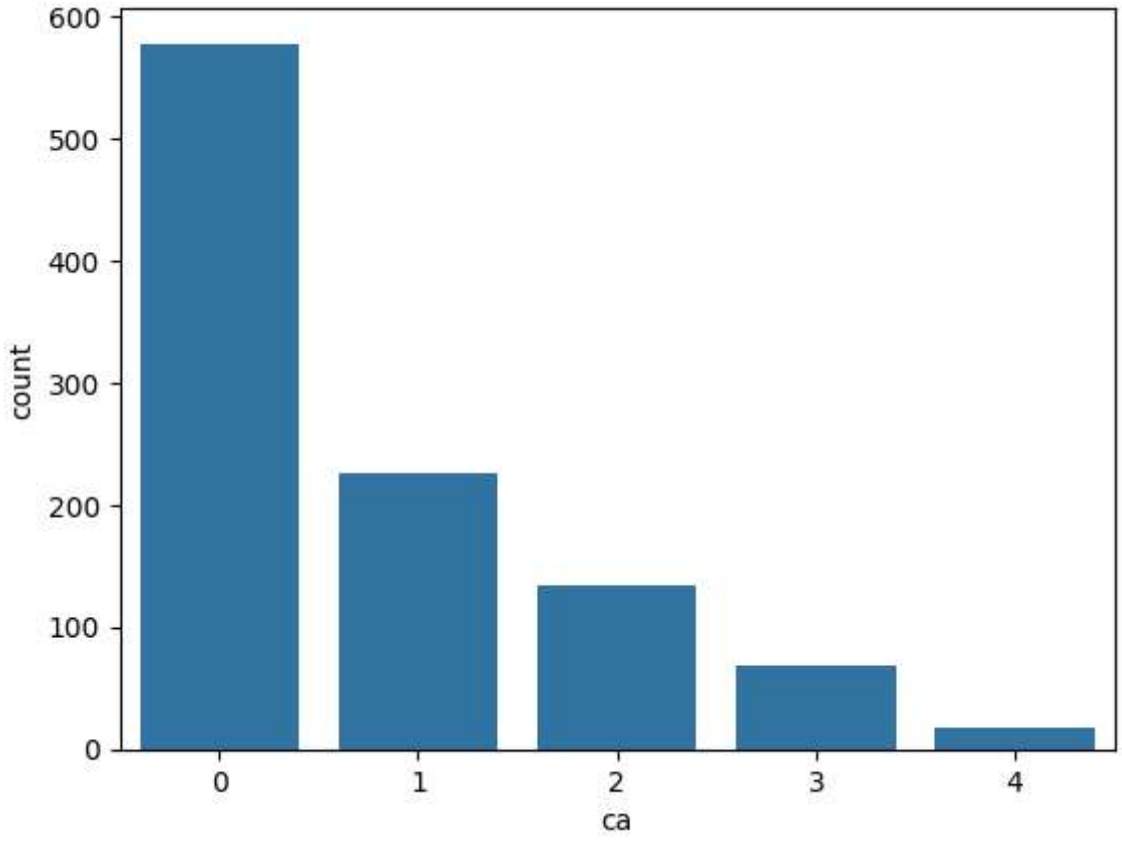
In [28]:
```
sns.countplot(x=df["sex"])
```

Out[28]:
```
<Axes: xlabel='sex', ylabel='count'>
```

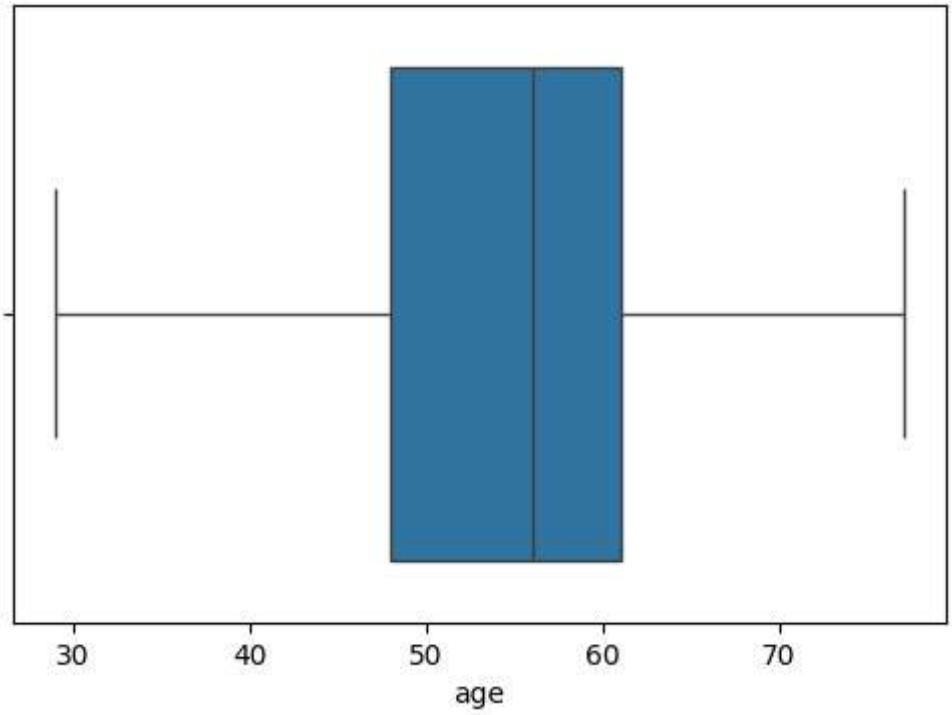


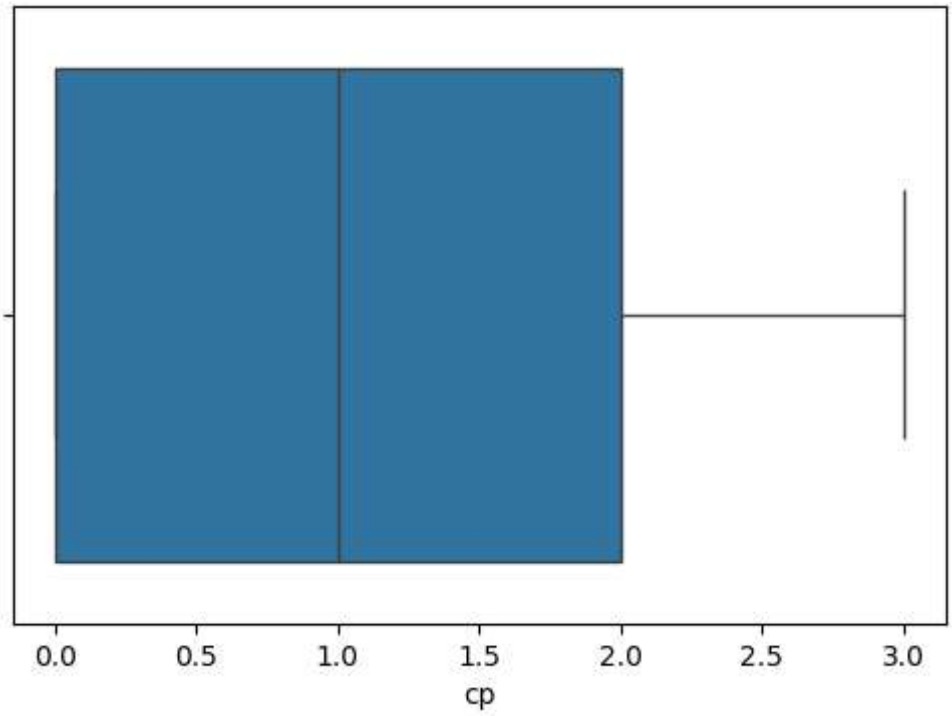
In [29]:
```
sns.countplot(x=df["ca"])
```

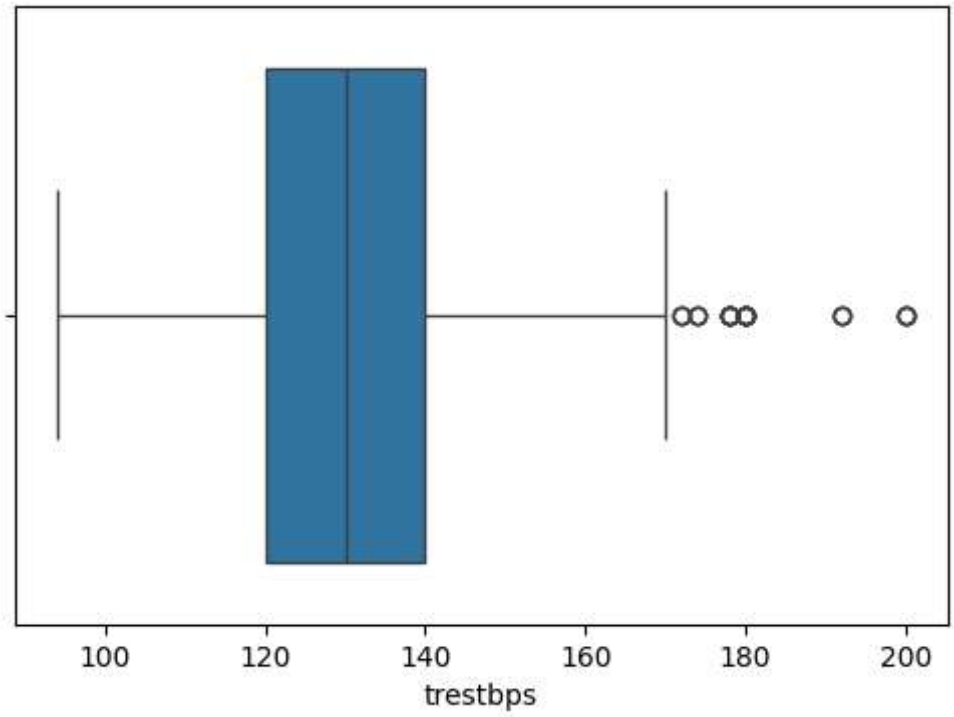
Out[29]: <Axes: xlabel='ca', ylabel='count'>

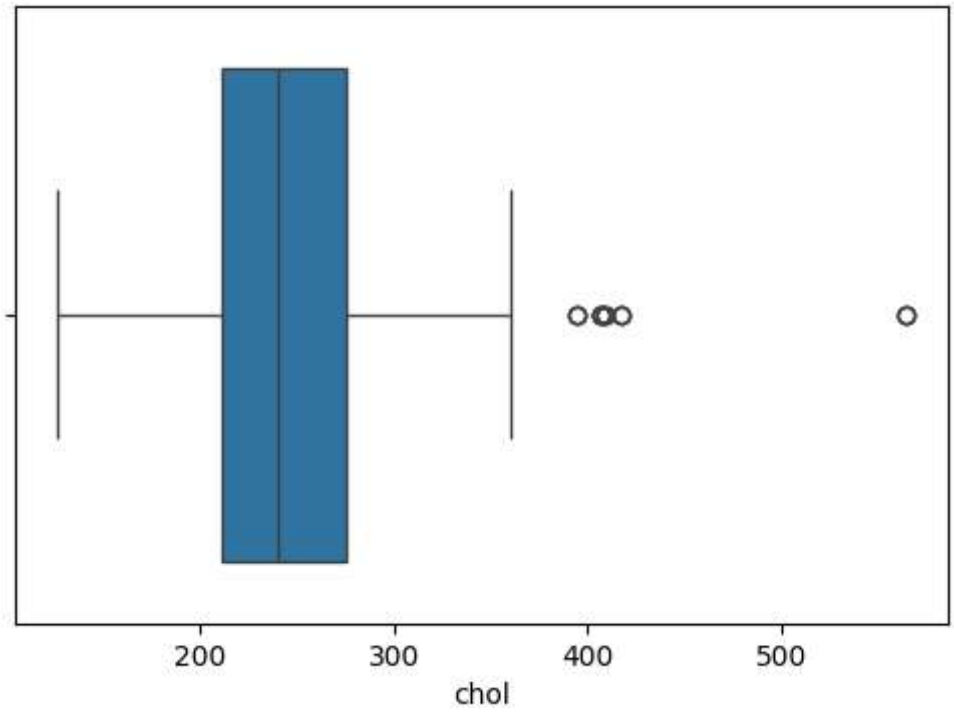


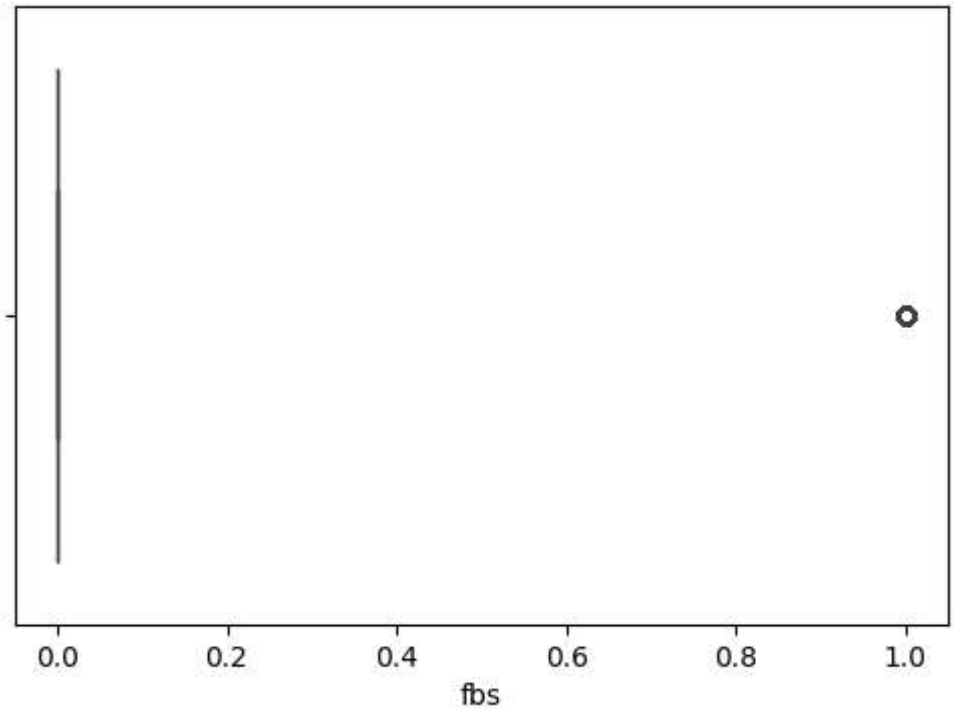
In [31]:
```
for col in numeric_columns:
    plt.figure(figsize=(6,4))
    sns.boxplot(x=df[col])
```

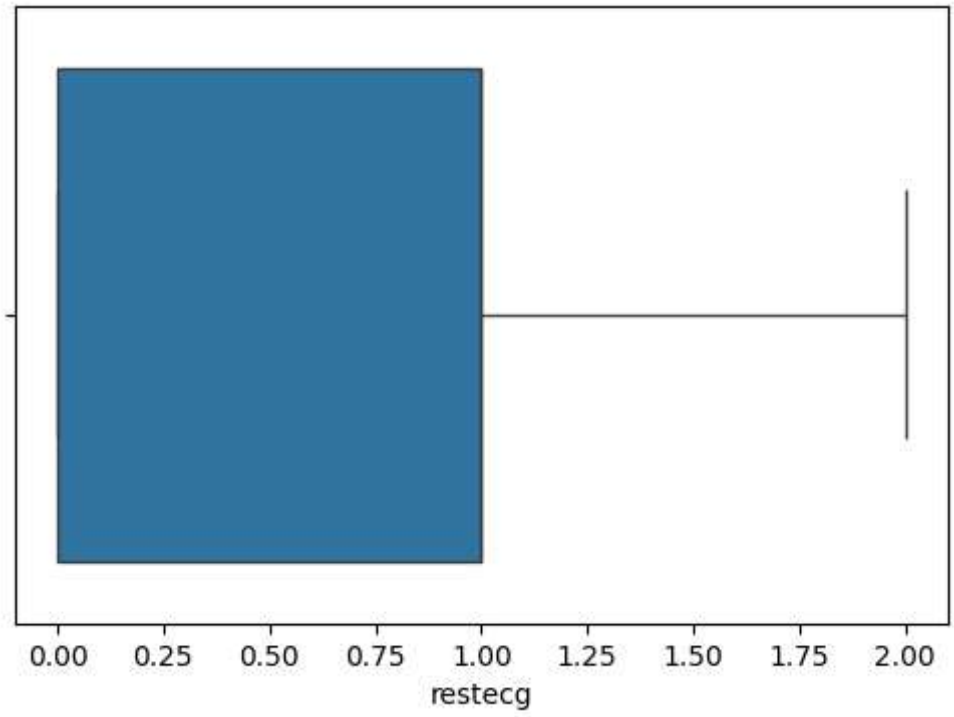


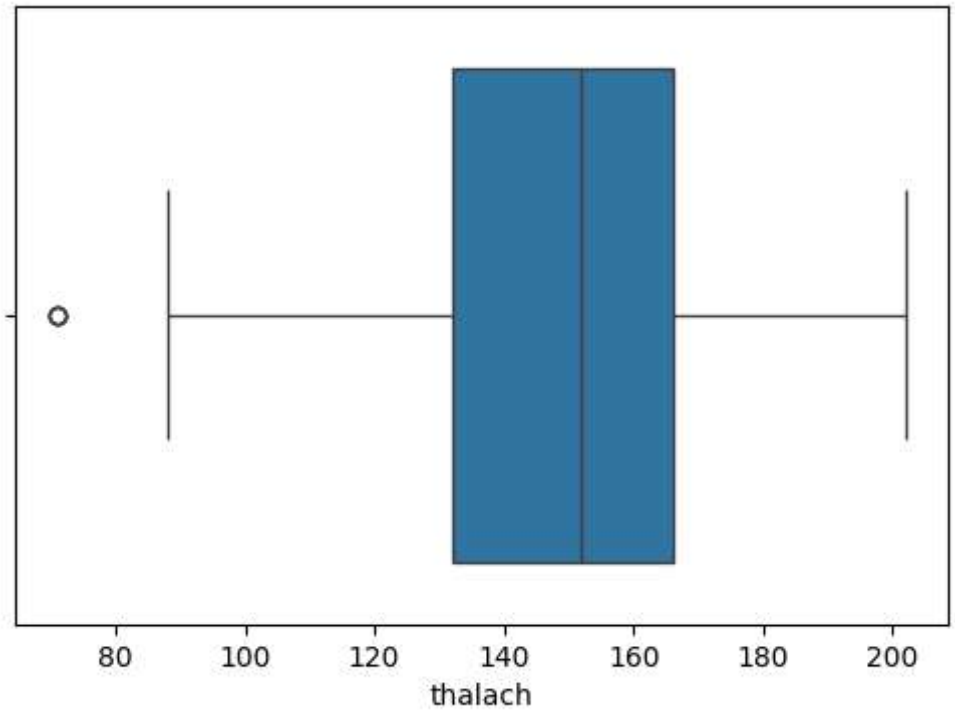


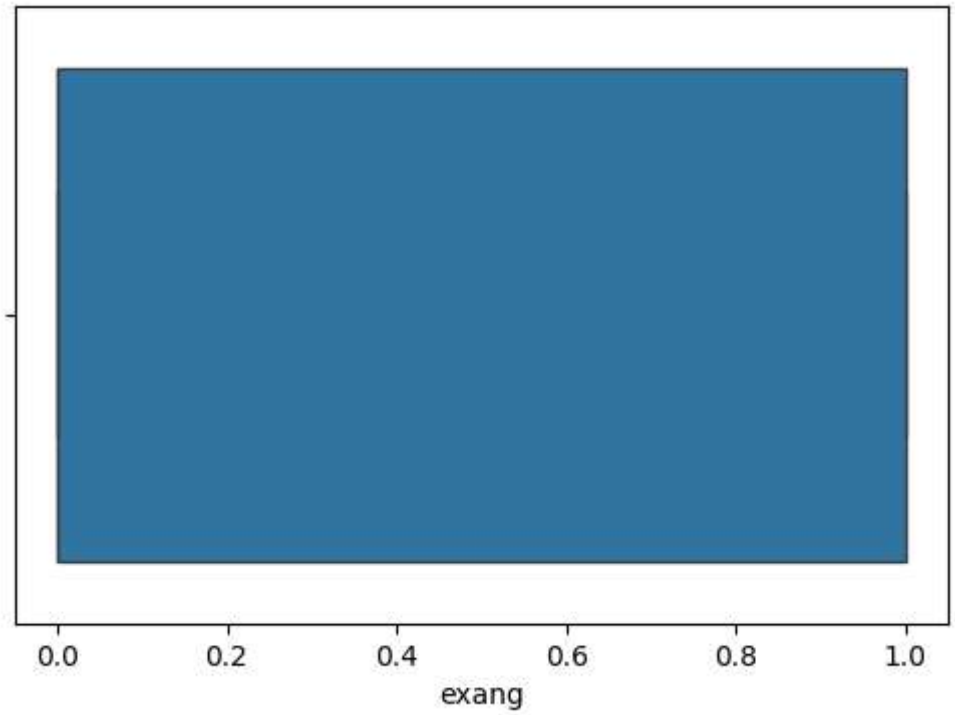

100
120
140
160
180
200
trestbps

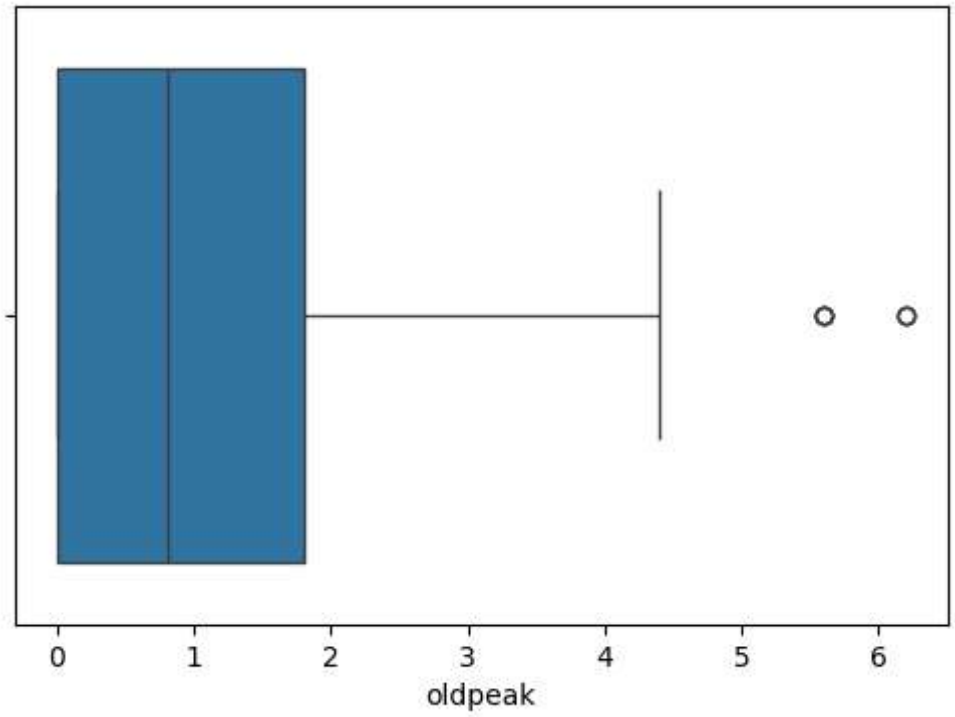

200
300
400
500
chol

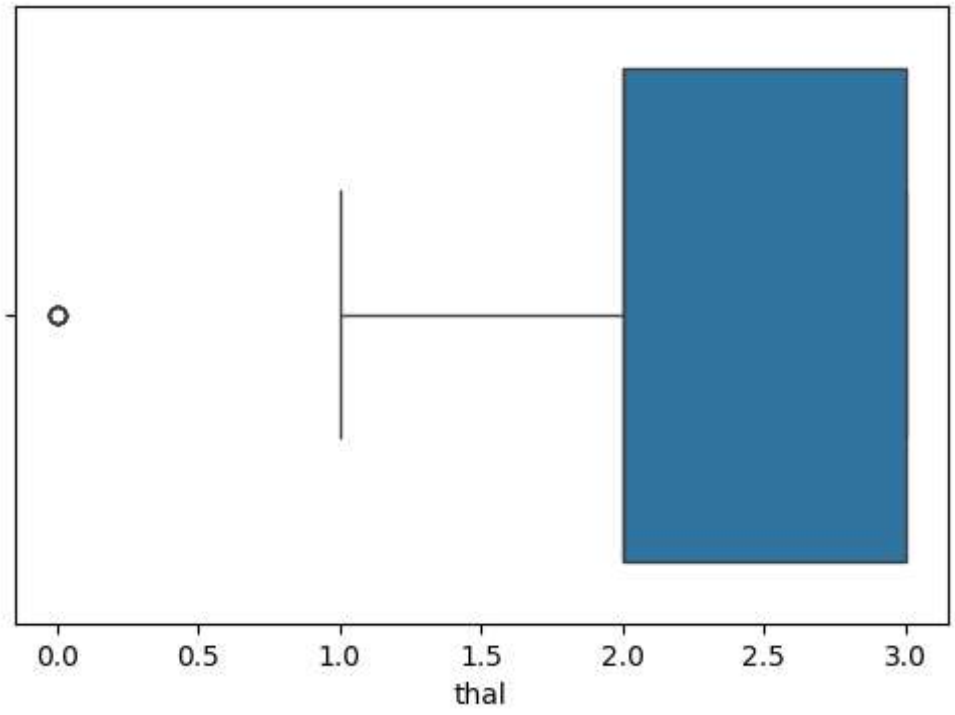

0.0
0.2
0.4
0.6
0.8
1.0
fbs

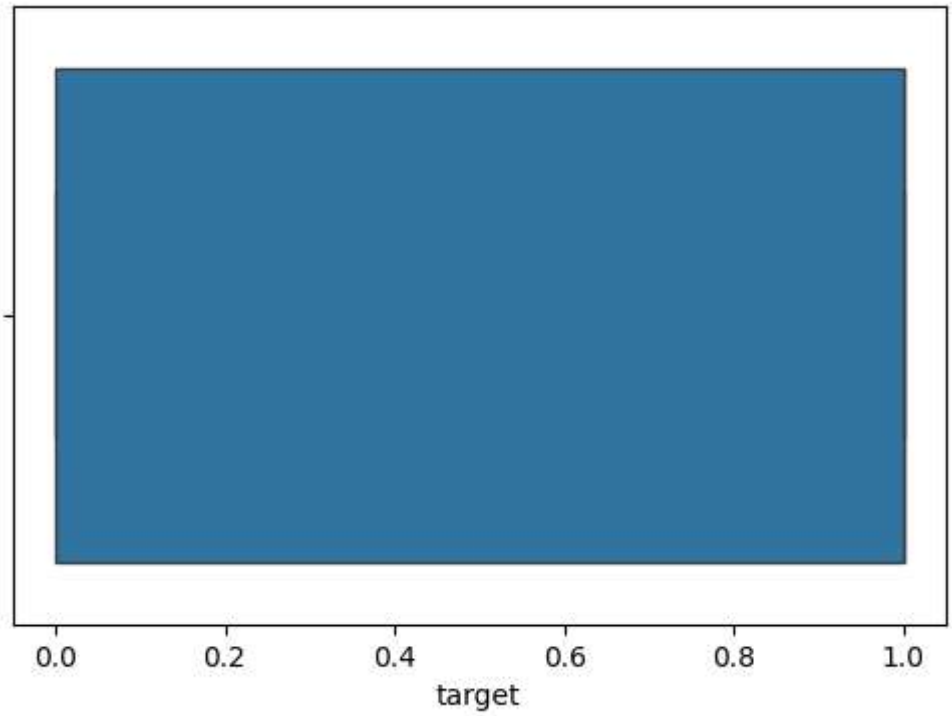

0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
restecg


80
100
120
140
160
180
200
thalach


0.0
0.2
0.4
0.6
0.8
1.0
exang

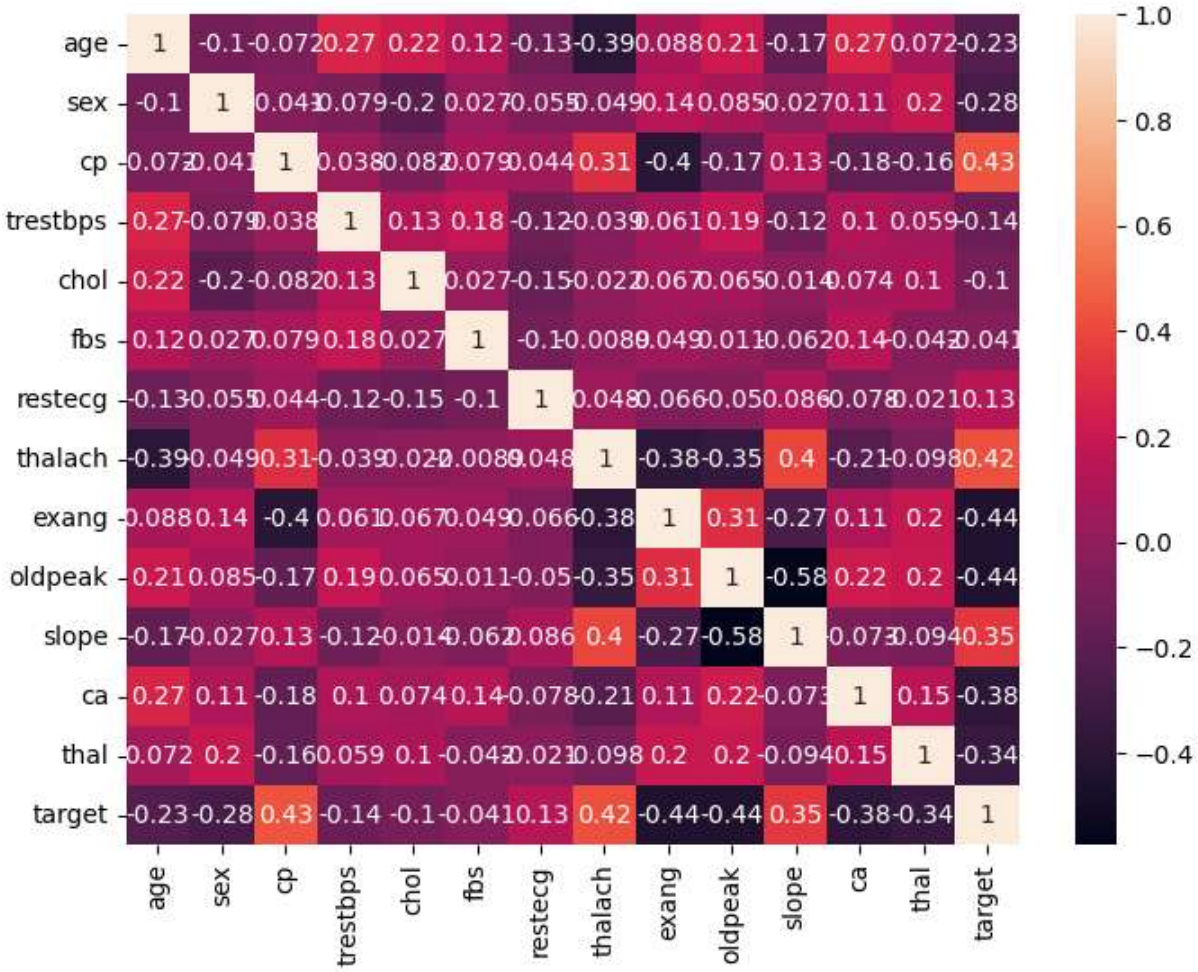

0
1
2
3
4
5
6
oldpeak


0.0
0.5
1.0
1.5
2.0
2.5
3.0
thal



In [34]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True),annot=True)
```

Out[34]: <Axes: >



Data Cleaning and preprocessing

In [35]:
```
df_cleaned=df.copy()
```

In [36]:
```
df_cleaned.head()
```

Out[36]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	t
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	

In [37]:
```
df_cleaned.shape
```

Out[37]:
```
(1025, 14)
```

In [38]:
```
df_cleaned.drop_duplicates(inplace=True)
```

In [39]:
```
df_cleaned.shape
```

Out[39]:
```
(302, 14)
```

In [40]:
```
df_cleaned.isnull().sum()
```

Out[40]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

In [41]:
```
df_cleaned.dtypes
```

Out[41]:
```
age           int64
sex           int64
cp            int64
trestbps      int64
chol          int64
fbs           int64
restecg       int64
thalach       int64
exang         int64
oldpeak     float64
slope         int64
ca            int64
thal          int64
target        int64
dtype: object
```

In [42]:
```
df_cleaned.head()
```

Out[42]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	t
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	

In [43]:
```
df_cleaned.rename(columns={
    "sex" :"is_male",
    "exang" :"is_exang"},inplace=True)
```

In [44]:
```
df_cleaned.head()
```

Out[44]:

	age	is_male	cp	trestbps	chol	fbs	restecg	thalach	is_exang	oldpeak	slope	ca	t
0	52	1	0	125	212	0	1	168	0	1.0	2	2	
1	53	1	0	140	203	1	0	155	1	3.1	0	0	
2	70	1	0	145	174	0	1	125	1	2.6	0	0	
3	61	1	0	148	203	0	1	161	0	0.0	2	1	
4	62	0	0	138	294	1	1	106	0	1.9	1	3	

In [ ]: